

TRADE TRACK – AN INTELLIGENT MARKETPLACE MANAGEMENT SYSTEM

*A Project Report submitted in partial fulfillment of the requirements for
the award of the degree of*

Master of Computer Applications

By

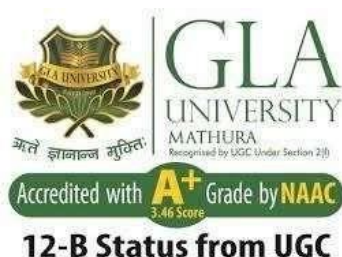
Mohit Verma	2484200119
Ankit Kumar	2484200028
Rahul Yadav	2484200153
Tinku Sharma	2484200197

Group No. 51
under the Guidance of

Mr. Navyum

Department of Computer Engineering & Applications

Institute of Engineering & Technology



GLA University
Mathura- 281406, INDIA
April, 2026



12-B Status from UGC

Department of Computer Engineering and Applications GLA University, 17
km Stone, NH#2, Mathura-Delhi Road, P.O. Chaumunhan, Mathura-281406
(U.P.)

Declaration

We hereby declare that the work is being presented in the M.C.A. Project “**Trade Track – An Intelligent Marketplace Management System**” in partial fulfillment of the requirements for the award of the *Master of Computer Applications* and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of our own work carried under the supervision of **Mr. Navyum, Dept. of CEA of Computer Engineering Department**).

Sign_____

Name of Student: Mohit Verma

University Roll no: 2484200119

Sign_____

Name Of Student: Ankit Kumar

University Roll no: 2484200028

Sign_____

Name of Student: Rahul Yadav

University Roll no: 2484200153

Sign_____

Name Of Student: Tinku Sharma

University Roll no: 2484200197

Certificate

This is to certify that the above statements made by the candidate are correct to the best of my/our knowledge and belief.

Supervisor
(Mr. Navyum)
Designation of Supervisor
Dept. of Computer Engg, & App.

Project Co-ordinator
(Dr. Sachendra Singh Chauhan)
Assistant Professor
Dept. of Computer Engg, & App.

Program Co-ordinator
(Mr. Siddharth Singh)
Assistant Professor
Dept. of Computer Engg, & App

Acknowledgement

The satisfaction accompanying the successful completion of the project is incomplete without mentioning a few names. We take this opportunity to acknowledge the efforts of the many individuals who helped us to make this project possible.

Firstly, we would like to express our heartfelt appreciation and gratitude to our project guide **Mr. Navyum, Dept. of Computer Engineering & Applications Department**. Her vision and execution aimed at creating a structure, definition, and realism around the project and fostered the ideal environment for us to learn and do. Despite his busy schedule, this project results from her teaching, encouragement, and input in the numerous meetings she had with us.

The experience was a novel one and we would like to thank all the people, who have lent their valuable time for the completion of the report. Without their consideration, it would have been difficult to complete the report

Sign_____

Name of Student: Mohit Verma

University Roll no: 2484200119

Sign_____

Name Of Student: Ankit Kumar

University Roll no: 2484200028

Sign_____

Name of Student: Rahul Yadav

University Roll no: 2484200153

Sign_____

Name Of Student: Tinku Sharma

University Roll no: 2484200197

Abstract

This Trade Track – An Intelligent Marketplace Management System is a full-stack web application that allows users to aims to provide a unified and intelligent SaaS-based platform that helps startups and small-to-medium enterprises (SMEs) manage their complete business operations in one place. Today, most businesses rely on multiple separate tools for Marketing, CRM, HR management, Billing, Inventory tracking, and Project coordination. This results in data fragmentation, higher costs, workflow delays, and lack of real-time insights. It supports secure user registration and login with email verification, JWT-based authentication, and password hashing using bcrypt. All booking and user data is stored in MongoDB using Mongoose, with flexible schemas to handle different booking types and their specific details. Real-time seat and availability updates are implemented using Socket.IO so that multiple users can see live changes when seats are locked, confirmed, or cancelled. The frontend is built with React, TypeScript, and Tailwind CSS, providing a responsive and user-friendly interface with multi-language and multi-currency support.

In addition to these features, the system is designed with a modular architecture that separates frontend, backend, and database layers for better scalability and maintainability. It also includes proper validation mechanisms to ensure that users enter correct and complete information during Startup. By integrating these modules within one centralized platform, Trade Track reduces the need for multiple costly software tools and creates a streamlined workflow for businesses. The system also offers real-time data access, smart analytics, and a user-friendly dashboard that supports better decision-making.

Contents

Declaration Certificate

Acknowledge Abstract

List of figures

CHAPTER 1 Introduction

- 1.1 Overview and Motivation
- 1.2 Objective
- 1.3 Summary of Similar Application

CHAPTER 2 Software Requirement Analysis

- 2.1 Requirement Analysis
 - 2.1.1 Functional Requirement
 - 2.1.2 Non-Functional Requirements
- 2.2 Feasibility Analysis
 - 2.2.1 Technical Feasibility
 - 2.2.2 Economic Feasibility
- 2.3 Module Description
- 2.4 Use Case Description

CHAPTER 3 Software Design

- 3.1 System Architecture
 - 3.1.1 User Interface Layer
 - 3.1.2 Application Logic Layer
- 3.2 Data Flow Design
 - 3.2.1 User Data Flow
 - 3.2.2 Admin Data Flow

- 3.3 System Architectural Diagram
- 3.4 UML Diagrams
 - 3.4.1 Class Diagram
 - 3.4.2 Summary

CHAPTER 4 Implementation and User Interface

- 4.1 Implementation Overview
- 4.2 Frontend Implementation
 - 4.2.1 User Interface Modules
 - 4.2.2 UI Components
 - 4.2.3 Sample UI Features
- 4.3 Backend Implementation
 - 4.3.1 Backend Modules Implemented
- 4.4 Database Implementation
 - 4.4.1 Key Collections/Tables
- 4.5 Module Integration
- 4.6 User Interface Screens
 - 4.6.1 Login Page
 - 4.6.2 Admin Dashboard
 - 4.6.3 CRM Interface
 - 4.6.4 Project & Task Management Screen
 - 4.6.5 Billing Screen
- 4.7 Implementation Challenges

CHAPTER 5 Software Testing

- 5.1 Introduction to Software Testing
- 5.2 Types of Testing Performed
 - 5.2.1 Unit Testing
 - 5.2.2 Integration Testing
 - 5.2.3 System Testing

- 5.2.4 Functional Testing
- 5.2.5 Non-Functional Testing
- 5.2.6 System Testing
- 5.2.7 User Acceptance Testing (UAT)

- 5.3 Test Cases
 - 5.3.2 CRM Module Test Cases
 - 5.3.3 Project & Task Management Test Cases
 - 5.3.4 Billing & Invoice Test Cases
 - 5.3.5 Inventory Module Test Cases
- 5.4 Bug Reporting & Resolution
- 5.5 Test Results

CHAPTER 6 Conclusion

- 6.1 Summary
- 6.2 Future Scope

REFERENCE

List of Figures

1. **Figure 3.2.1.1:** Level 0 DFD
2. **Figure 3.2.1.2:** Level 1 DFD
3. **Figure 3.2.1.3:** Level 2 DFD
4. **Figure 3.2.2:** User-Side Use Case Diagram_
5. **Figure 3.2.3:** Admin-Side Use Case Diagram
6. **Figure 3.2.4:** System Architectural Diagram
7. **Figure 3.3.1:** Class Diagram
8. **Figure 3.3.2:** Sequence Diagram
9. **Figure 3.3.3:** Object Diagram
10. **Figure 3.4.1:** ER Diagram

Chapter 1

Introduction

The **Tarde Track – An Intelligent Marketplace Management System**. In today's competitive business environment, startups and small-to-medium enterprises (SMEs) rely on multiple separate tools to handle their daily operations—such as marketing, customer relationship management, project coordination, billing, staff management, and inventory tracking. While each tool may work well individually, using many disconnected platforms often creates major challenges for businesses. Data becomes scattered, workflows become slow, and teams struggle to maintain proper coordination. As a result, business owners face difficulties in monitoring performance, making timely decisions, and maintaining overall operational efficiency.

A common issue observed in real-world SMEs is data fragmentation. Customer information may be stored in one tool, project details in another, billing records somewhere else, and inventory reports in a completely different system. This lack of centralization leads to confusion, manual errors, and delays. Many businesses also struggle with the cost of maintaining multiple subscriptions for CRM systems, HR tools, billing software, and inventory platforms. The absence of real-time insights further limits their ability to improve performance or identify business opportunities. Moreover, employees often waste time switching between tools, affecting productivity and increasing operational complexity.

These challenges highlight the need for a unified, intelligent, and affordable business management platform that brings all essential functions together in one place. The motivation behind developing “Trade Track – An Intelligent Marketplace Management System” comes from observing the real day-to-day difficulties faced by startups and SMEs. The platform aims to offer a single operating system for businesses, integrating Marketing & CRM, Internal Workflow Management, Staff & Billing, and Inventory & Analytics into one seamless, cloud-based solution. By doing so, it eliminates the inefficiencies caused by tool fragmentation and creates a smooth workflow for every business department. Trade Track focuses on providing real-time data visibility, automation, and smart analytics to help business owners make informed, data-driven decisions. With features like centralized dashboards, automated processes, integrated communication, and secure cloud storage, the system offers a modern, scalable, and user-friendly solution. Built using technologies such as Node.js, Express.js, React.js, and cloud infrastructure, Trade Track ensures high performance.

1.1 Overview and Motivation

The primary objective of this project is to develop a unified, intelligent, and cloud-based platform that simplifies, automates, and manages complete business operations for startups and small-to-medium enterprises (SMEs). The key aims of the Trade Track system are as follows:

1. To integrate all major business functions into a single platform: The system will combine Marketing & CRM, Internal Management, Staff & Billing, and Inventory & Analytics into one centralized solution, eliminating the need for multiple disconnected tools.
2. To streamline marketing and customer relationship management: Trade Track will help businesses track leads, manage customer interactions, run marketing campaigns, and monitor the sales funnel efficiently from one interface.
3. To improve internal workflow and project management :The platform will allow teams to manage projects, assign tasks, track progress, and collaborate seamlessly to enhance productivity and reduce delays.
4. To automate staff management and billing processes: Trade Track will provide features for employee records, attendance, payroll generation, client invoicing, and payment tracking—reducing manual errors and saving time.
5. To support inventory management and advanced analytics: The system will offer real-time stock updates, supplier management, and business intelligence dashboards to help businesses make data-driven decisions. (This module can be expanded in future versions.)
6. To deliver real-time insights and actionable data: Business owners and teams will have access to live dashboards, reports, and analytics that boost transparency and support better decision-making.
7. To ensure scalability, accessibility, and cloud-based performance: By deploying the platform on cloud infrastructure, the system ensures high availability, security, remote access, and the ability to scale as the business grows.

1.2 Objective

The primary objective of this project is to develop a unified, intelligent, and cloud-based platform that simplifies, automates, and manages complete business operations for startups and small-to-medium enterprises (SMEs).

The key aims of the Trade Track system are as follows:

- 1. To integrate all major business functions into a single platform:** The system will combine Marketing & CRM, Internal Management, Staff & Billing, and Inventory & Analytics into one centralized solution, eliminating the need for multiple disconnected tools.
- 2. To streamline marketing and customer relationship management:** Trade Track will help businesses track leads, manage customer interactions, run marketing campaigns, and monitor the sales funnel efficiently from one interface.
- 3. To improve internal workflow and project management:** The platform will allow teams to manage projects, assign tasks, track progress, and collaborate seamlessly to enhance productivity and reduce delays.
- 4. To automate staff management and billing processes:** Trade Track will provide features for employee records, attendance, payroll generation, client invoicing, and payment tracking—reducing manual errors and saving time.
- 5. To support inventory management and advanced analytics:** The system will offer real-time stock updates, supplier management, and business intelligence dashboards to help businesses make data-driven decisions. (This module can be expanded in future versions).
- 6. To deliver real-time insights and actionable data:** Business owners and teams will have access to live dashboards, reports, and analytics that boost transparency and support better decision-making.
- 7. To ensure scalability, accessibility, and cloud-based performance:** By deploying the platform on cloud infrastructure, the system ensures high availability, security, remote access, and the ability to scale as the business grows.
- 8. To reduce operational costs and complexity:** By replacing multiple paid software subscriptions with one unified system, Trade Track makes business management more affordable and simplifies daily workflows.

These objectives collectively aim to modernize business operations, improve efficiency, reduce manual effort, and empower SMEs with a smart, integrated, and easy-to-use business management platform.

1.3 Summary of Similar Applications

Several businesses today use a combination of separate tools and software to manage different parts of their operations. Popular platforms such as **Zoho, HubSpot, Odoo, Tally, QuickBooks, and separate CRM or HR tools** are commonly used by startups and SMEs. However, these systems come with several limitations when compared to the unified, intelligent, and fully integrated solution we aim to develop.

Most CRM or marketing tools focus only on lead management and customer communication. They do not provide internal project management, billing, or inventory features. Similarly, accounting and billing software like Tally or QuickBooks handle only financial records and invoicing, but do not support marketing, HR, or workflow automation. Inventory tools also work independently and do not sync with sales, projects, or staff updates. As a result, businesses often need 5–7 different tools to manage everyday operations, which leads to confusion, high costs, and scattered data. ERP systems like Odoo or Zoho One attempt to combine multiple modules, but they come with challenges such as high subscription costs, complex setup, and limited flexibility for small businesses. Many SMEs either cannot afford them or find them too complicated for their needs. Free or low-cost alternatives lack advanced features, real-time analytics, cloud scalability, and integrated dashboards. In many small businesses, operations are still handled through **Excel sheets, Google Forms, WhatsApp groups, and offline documentation**. This approach is time-consuming, prone to errors, and lacks automation. Team members often miss important updates, and business owners struggle to get a clear picture of overall performance. Without a unified system, tracking sales, projects, staff activities, or inventory becomes difficult and inefficient. Compared to all these existing tools, Trade Track offers a unique advantage by integrating Marketing & CRM, Internal Management, Staff & Billing, and Inventory & Analytics into a single cloud-based system. It combines automation, real-time insights, centralized data, and user-friendly dashboards—features that are not available together in any currently used solution. This makes Trade Track a modern, scalable, and highly impactful platform tailored specifically for startups and SMEs seeking an all-in-one business management ecosystem

Chapter 2

SOFTWARE REQUIREMENT ANALYSIS

The success of a software project depends heavily on clear understanding and analysis of requirements before development begins. Proper requirement analysis ensures that the system behaves as expected, supports all business functions, and meets user needs efficiently. This chapter explains the requirements, feasibility analysis, modules, use cases, and system architecture for Trade Track – An Intelligent Marketplace Management System.

2.1 Requirement Analysis

Requirement analysis is the process of identifying **what** the system should do and **how** it should perform. It includes both:

- **Functional Requirements** (What features the system must provide)
- **Non-Functional Requirements** (How the system should behave)

2.1.1 Functional Requirements

1. User Registration & Login (Role-Based Access):

- Business Owners, Employees, and Admins can register and log in securely.
- Role-based dashboards are shown depending on user type (Owner, Staff, Accountant, Inventory Manager, etc.).

2. Marketing & CRM Module:

- Add and track leads/customers.
- Manage marketing campaigns.
- Update customer interaction history.
- Track sales funnel stages (Lead → Prospect → Customer).

3. Internal Management (Projects & Tasks):

- Create projects, assign tasks, set deadlines.
- Team members can update task progress.
- Dashboard shows ongoing, completed, and delayed tasks.

4. Staff Management:

- Add employee profiles.
- Manage attendance and working hours.
- Process payroll and generate salary slips.

5. Billing & Invoicing:

- Create invoices for clients.
- Track payments, due amounts, and transaction history.
- Download or email invoices directly.

6. Inventory Management (Future Expansion):

- Add products, update stock quantity.
- Low-stock alerts to the manager.
- Supplier management and purchase tracking.

7. Analytics & Reporting:

- Real-time reports for sales, revenue, projects, and employee performance.
- Dashboard visualizations for key business metrics.
- Export reports as PDF/Excel.

8. Notification System:

- Send updates for pending tasks, payment reminders, project deadlines.
- Email/SMS integration in future versions.

2.1.2 Non-Functional Requirements

1. Performance

- Should respond to user actions within 2–3 seconds.
- Capable of handling multiple users simultaneously.

2. Scalability

- Cloud architecture supports growing number of clients, employees, and business data.

3. Security

- Encrypted passwords (bcrypt/JWT).
- Role-based permission control.
- Secure data handling and API authentication.

4. Usability

- Simple, intuitive UI suitable for business users with minimal technical knowledge.

5. Reliability

- System should operate smoothly without crashes.
- Cloud backup ensures no data loss.

2.2 Feasibility Analysis

A feasibility study determines whether the project is practical and can be successfully developed.

2.2.1 Technical Feasibility

- Tools required: Node.js, Express.js, React.js, MongoDB/MySQL, REST APIs, Cloud Deployment platforms (AWS/Azure/GCP).
- Easy to develop using modern web technologies.
- Supports microservices and scalable cloud architecture.

2.2.2 Economic Feasibility

- Most technologies used are open-source.
- Minimal hardware and software costs.
- Affordable for startups and SMEs.

2.2.3 Operational Feasibility

- User-friendly dashboards for all roles (Owner, Staff, Accountant, etc.).
- Automated workflows reduce manual effort.
- Easy for non-technical users to operate.

2.3 Module Description

The proposed Trade Track system is divided into major functional modules. Each module handles specific business operations and interacts smoothly with others.

1. User Module (Owner/Employee/Manager):

- **1.1 Register / Login:** Working Mechanism:
- User registers with name, email, and password.
- System validates user details and encrypts passwords.
- Upon login, role-based dashboard is shown.

2. Marketing & CRM Module:

- Users add leads with contact details.
- CRM tracks stages: New Lead → Follow-Up → Converted.
- Notes, calls, and follow-up reminders are stored.
- Campaign performance monitored via dashboard.

2.4 Use Case Descriptions:

Use Case 1: Add New Customer/Lead

- Actor: Owner/Marketing Executive
- Description: User adds customer details to CRM.
- Outcome: Customer added and sales funnel updated.

Use Case 2: Create Invoice

- Actor: Accountant/Owner
- Description: User generates invoice for a client.
- Outcome: Invoice saved in system and marked as paid/unpaid.

Use Case 3: Assign Project Task

- Actor: Manager
- Description: Assigns task to team member.
- Outcome: Task appears in employee dashboard.

Chapter 3

SYSTEM DESIGN

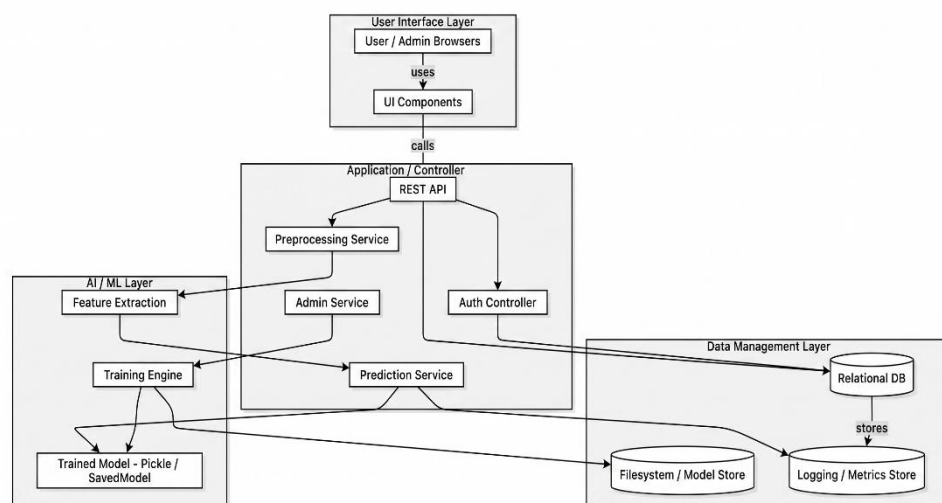
System design forms the foundation of the Trade Track – Intelligent Marketplace Management System, translating all conceptual and functional requirements into a structured, scalable, and implementable architecture. This chapter explains the system's architecture, data flow, module interactions, database structure, and UML diagrams. The purpose of this design is to ensure clarity, maintainability, reliability, and smooth system performance across all business operations such as CRM, Projects, Billing, Staff Management, and Inventory.

3.1 System Architecture

The system follows a three-tier cloud-based architecture, consisting of:

1. User Interface Layer
2. Application Logic Layer
3. Data Management Layer

Each layer works independently yet in coordination to deliver a seamless business management experience.



The User Interface Layer ensures that interactions are intuitive, responsive, and user-friendly, providing a seamless booking experience for all users.

3.1.1 User Interface Layer

This layer provides all user-facing screens for:

- Business Owners
- Employees (Marketing, HR, Accounts, Inventory Managers)
- Admin

The UI allows:

- Registration & Login
- CRM operations (Add leads, update customer status)
- Project & Task management
- Staff attendance & payroll actions
- Billing & invoice generation
- Inventory updates
- Dashboard views & analytics

The interface is designed to be clean, responsive, and user-friendly, ensuring that even non-technical business users can easily operate the system.

3.1.2 Application Logic Layer

This is the core processing layer that handles all business functions. It includes :

- CRM Controller
- Project & Task Management Module
- Staff & Payroll Processing
- Billing Controller
- Inventory Management Engine
- Notification & Reminder system
- Reporting & Analytics engine
- Role-Based Access Control (RBAC)
- API Communication Services

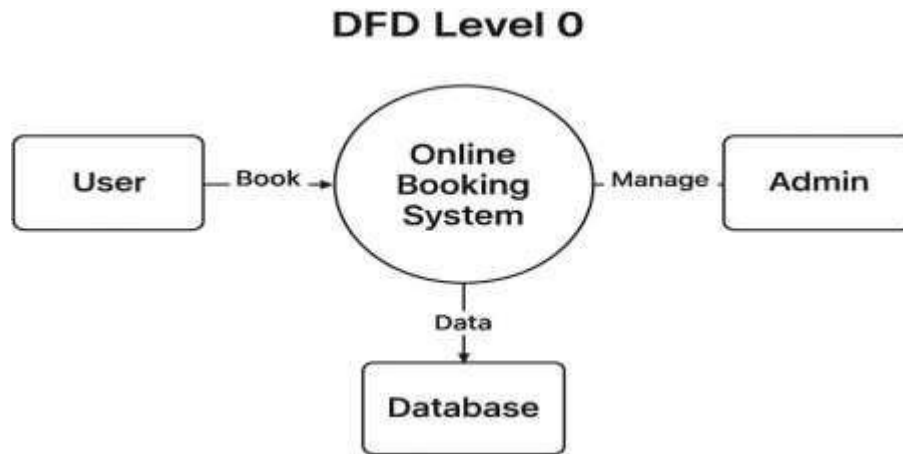


Figure 3.2.1.1: Level 0 DFD

3.1.3 Data Management Layer

This layer manages all essential business data, including:

- User, Employee, and Role records
- Customer and Lead details
- Project, Task, and Workflow data
- Attendance and Payroll information
- Client invoices & payment history
- Inventory items, suppliers, and stock records
- Reports and analytics logs

The database is structured using **MongoDB**, depending on deployment.

It ensures secure, fast, and reliable data storage.

3.2 Data Flow Design

Data flows through the system in an organized, step-by-step manner to ensure accuracy and efficiency.

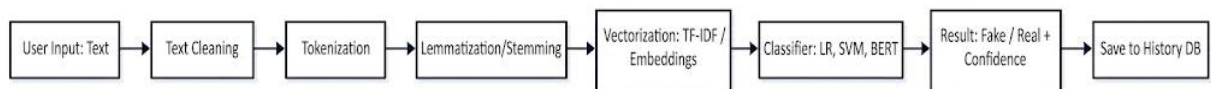
Level 0 DFD (Context Diagram)

The Level 0 DFD represents the entire Trade Track platform as a **single unified**

system.

External entities include:

- Business Owner
- Employees (CRM, HR, Accounts, Inventory)
- Admin

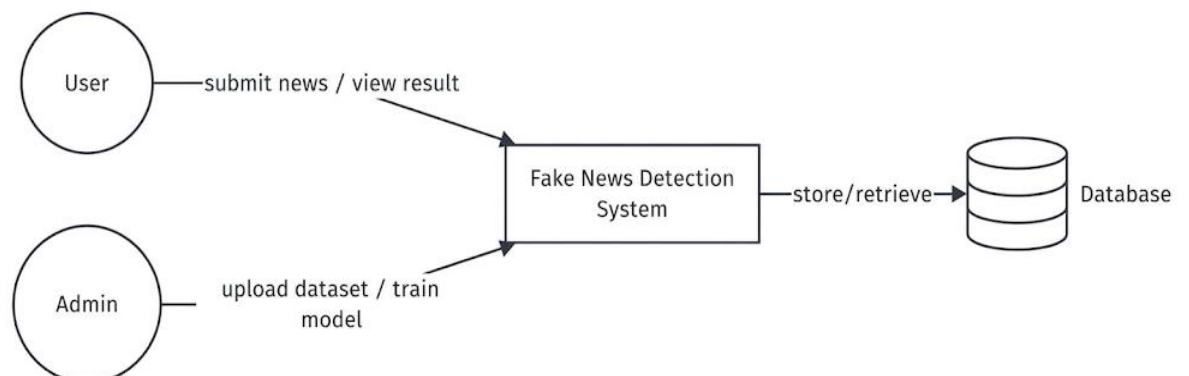


These entities interact with the central system to perform operations like CRM updates, billing, staff management, and project coordination.

Level 1 DFD – Explanation:

The Level 1 DFD breaks the system into major sub-processes such as:

1. User Authentication
2. CRM Operations (Customer/Lead management)
3. Project & Task Management
4. Staff & Payroll Processing
5. Billing & Invoicing
6. Inventory Management
7. Analytics & Reporting
8. Database Operations



Each of these modules interacts with the database for storing and retrieving

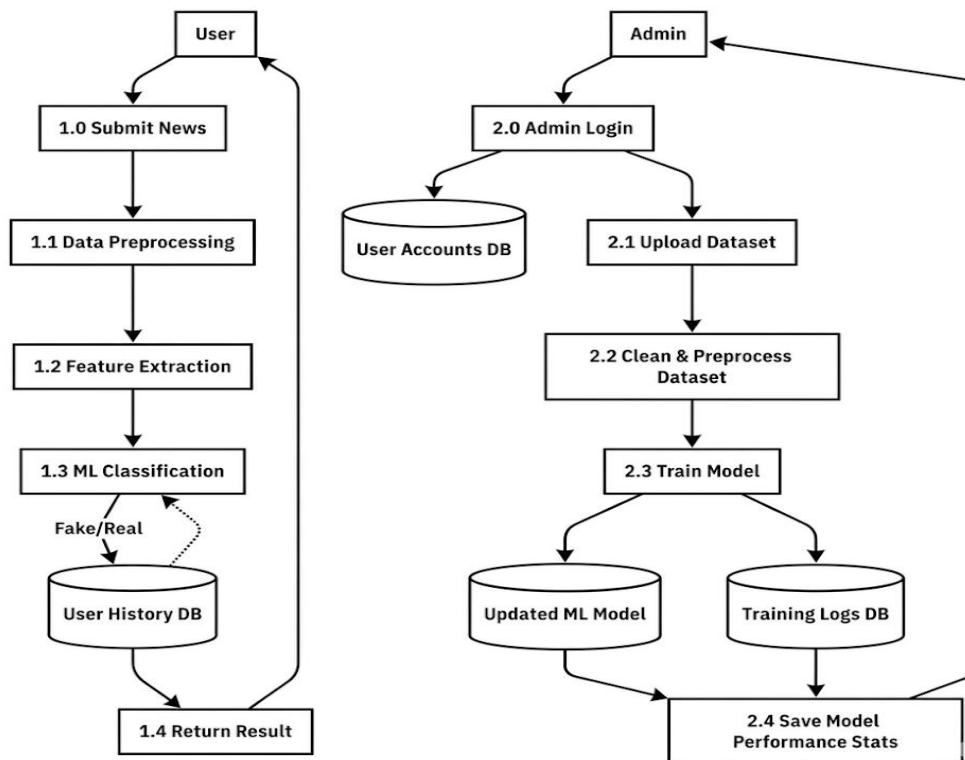
information.

Level 2 DFD – Explanation:

A more detailed breakdown is shown below:

1. User Authentication

- User enters credentials
- System validates credentials
- Role-based dashboard is displayed
- Login state is maintained using sessions/JWT



2. CRM (Marketing & Customer Management)

- User enters lead/customer information
- System validates details
- CRM pipeline updates stage (Lead → Prospect → Customer)
- Data stored in CRM tables

3. Project & Task Management

- Manager creates project + tasks
- Tasks assigned to employees
- Employees update progress
- Dashboard shows task completion status

4. Staff Management & Payroll

- HR updates attendance
- System calculates working hours
- Payroll generation and salary slip creation
- Records stored in database

5. Billing & Invoicing

- User creates client invoice
- System calculates taxes and totals
- Invoice saved + downloadable
- Payment status updated

6. Inventory Management

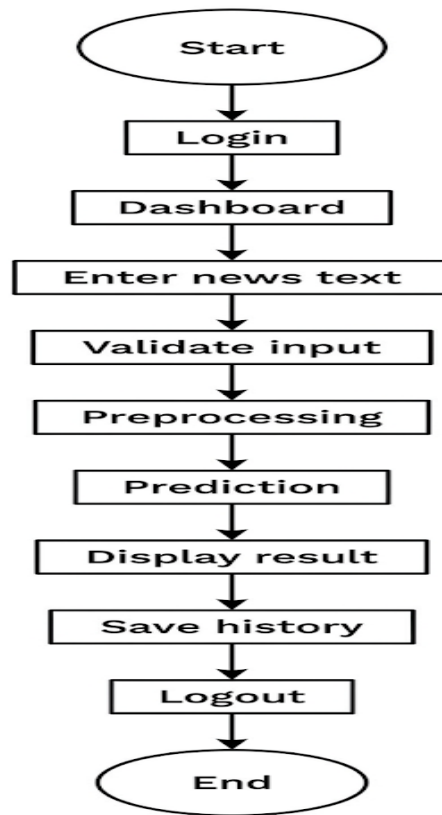
- Add/update stock
- Low-stock alert triggered
- Supplier records managed
- Purchase/sales stock adjusted

7. Analytics & Reporting

- System gathers live data
- Generates graphs and reports
- Visual dashboard shown to users

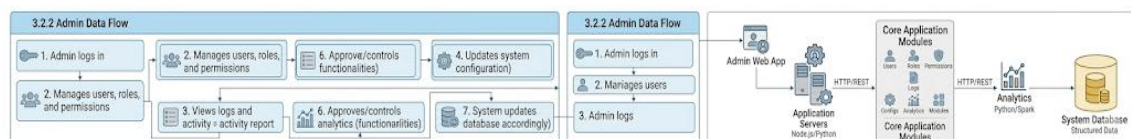
3.2.1 User Data Flow

1. User logs into system
 2. Accesses CRM, Billing, Projects, or Inventory
 3. Inputs data (customer info, invoices, tasks, etc.)
 4. Application Logic processes the data
 5. Data stored in the database
 6. Updated results returned to the user interface
 7. User views output (invoice, report, project status, CRM update)
-



3.2.2 Admin Data Flow

1. Admin logs in
2. Manages users, roles, and permissions
3. Views logs and activity report.
4. Updates system configuration
5. Monitors business analytics
6. Approves/controls module functionalities
7. System updates database accordingly

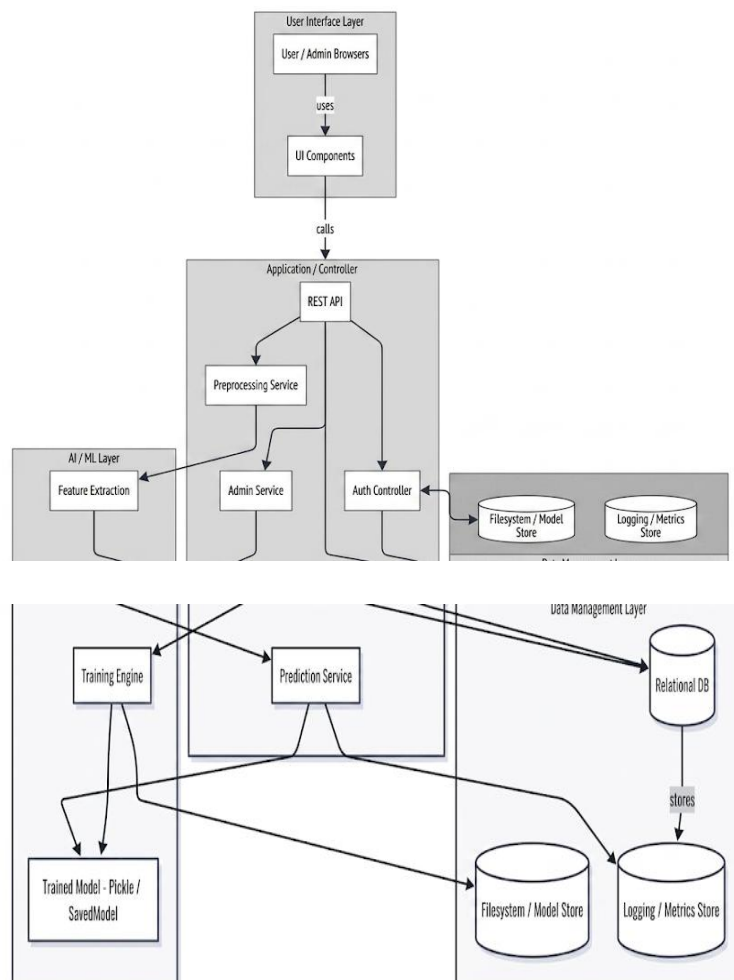


3.3 System Architectural Diagram

- User Interface Layer (Web/Mobile Screens)
- Controller Layer (Handles user requests, validations)

- Business Logic Layer (CRM engine, Payroll engine, Inventory engine, etc.)
- Database Layer (User, Customer, Staff, Stock, Invoice tables)
- Cloud Layer (Deployment, API hosting, file storage)
- This architecture ensures scalability, modularity, and clean separation of concerns.

3.5 UML Diagrams



3.4.1 Class Diagram

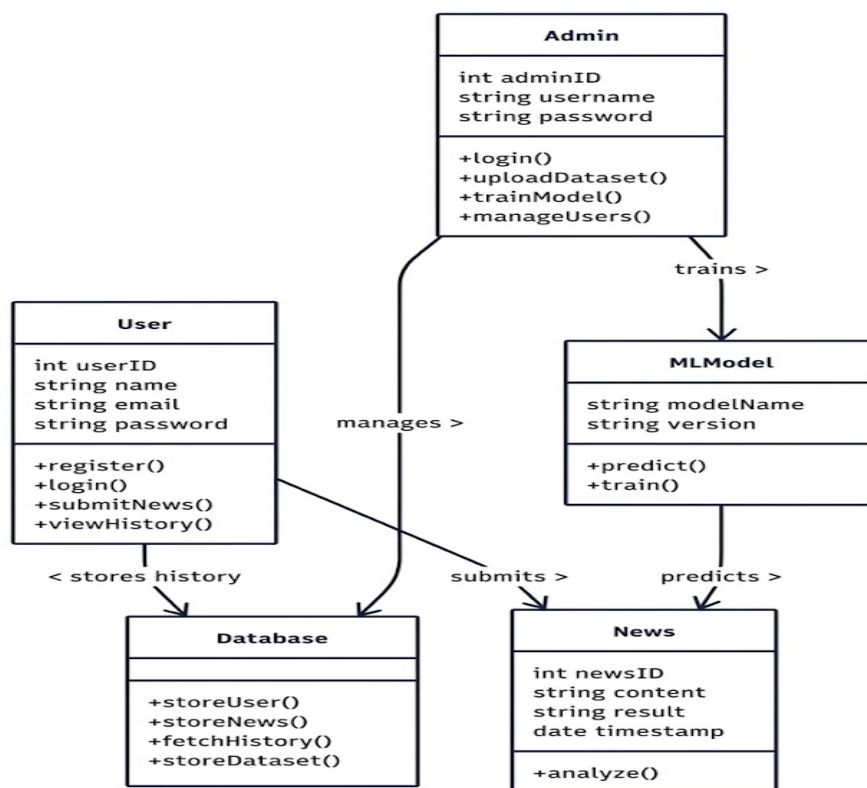
Main classes in Trade Track include:

- User
- Role

- Customer/Lead
- Project
- Task
- Employee
- Payroll
- Invoice
- Product/Inventory
- Report
- Admin

These classes interact through relationships such as:

- One-to-many (User → Tasks, Customer → Invoices)
- Many-to-one (Employee → Payroll)
- Many-to-many (Projects ↔ Employees)



The class diagram shows all attributes, methods, and relationships. Sequence Diagram

Sequence diagrams demonstrate time-based interactions such as:

1. Creating an Invoice

User → Invoice Module → System validates → Database stores → Invoice returned to

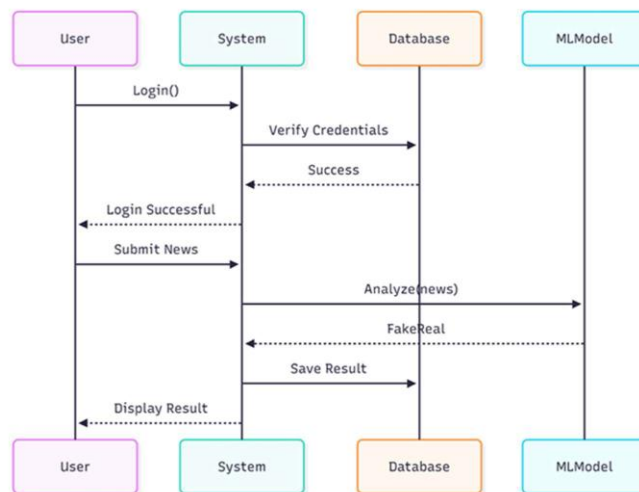
UI

2. Adding a New Customer

User inputs customer info → CRM Module → Database → Confirmation returned

3. Assigning a Task

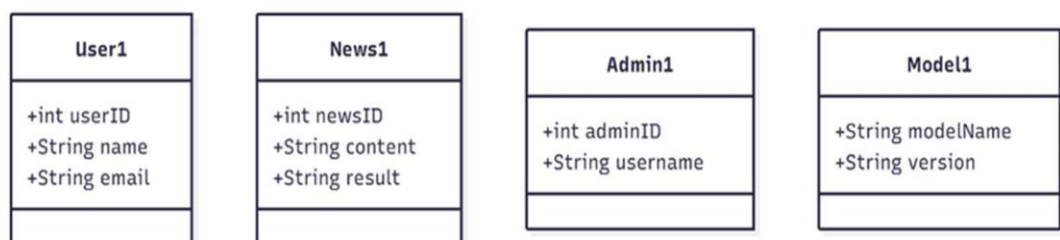
Manager → Task Module → Database → Employee notification



Object Diagram

Shows runtime instances such as:

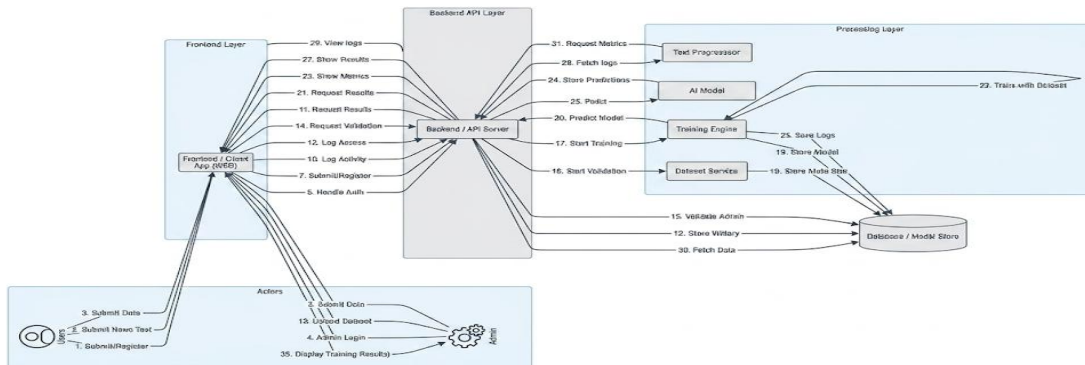
- User1
- Customer1
- Invoice54
- Project12
- StockItem3



This gives a snapshot of system state at a particular moment.

Collaboration Diagram: Shows how objects work together to complete tasks like:

- Generating payroll
- Updating inventory
- Moving customer through CRM pipeline



It highlights message paths and interactions.

E–R Diagram, Represents the database design for Trade Track.

Main entities include:

1. USER Table
2. ROLE Table
3. CUSTOMER / LEAD Table
4. PROJECT Table
5. TASK Table
6. EMPLOYEE Table
7. PAYROLL Table

Each table stores attributes such as:

- IDs (Primary keys)
- Foreign keys
- Business data fields

- Timestamps

This ER model forms the backbone of the database structure.

Chapter 4

Implementation and User Interface

This chapter explains the actual implementation of the **Trade Track** platform, including the technologies used, backend and frontend development, modulewise implementation flow, integration of functionalities, and the design of user interface screens. Implementation focuses on converting the system design into working software components, ensuring that each module performs reliably and interacts smoothly with

others.

The Trade Track system is implemented using **Node.js**, **Express.js** for backend APIs, **React.js** for the frontend, and **MongoDB/MySQL** for database storage. The architecture follows a modular pattern to ensure scalability, maintainability, and smooth updates.

4.1 Implementation Overview

The implementation of Trade Track is divided into the following major layers:

- 1. Frontend Implementation (User Interface)**
- 2. Backend Implementation (API and Business Logic)**
- 3. Database Implementation**
- 4. Module Integration & Workflow Execution**

Each layer performs specific tasks and collectively forms the complete system.

4.2 Frontend Implementation (User Interface)

The frontend of Trade Track is built using **React.js**, ensuring a clean, responsive, and user-friendly interface. Key features include:

4.2.1 User Interface Modules

The UI contains separate dashboards for:

- Admin
- Business Owner
- Employees (CRM, HR, Accounts, Inventory)

Each dashboard displays role-specific features.

4.2.2 UI Components

Common components used across the system include:

- Navigation bar
 - Side menu
 - Dashboard cards
 - Data tables
 - Forms for adding/modifying entries
 - Charts for analytics (using **Chart.js/Recharts**)
-

4.2.3 Sample UI Features

1. Login Screen

- Email and password fields
- "Forgot Password" option
- Role-based redirect after login

2. CRM Dashboard

- Add new leads/customers
- Track sales pipeline stages
- View customer history

3. Project Management Interface

- Create projects
- Assign tasks
- Track task progress visually

4. Billing & Invoice Screen

- Create, update, delete invoices
- Auto-calculate totals
- Display paid / unpaid status

4.3 Backend Implementation (Application Logic)

The backend is implemented using **Node.js (Express.js framework)**. Its primary responsibility is to handle:

- Business logic
- Data validation
- API routing
- Authentication
- Interaction with the database

4.3.1 Backend Modules Implemented

1. Authentication Module

- Password encryption using bcrypt
- JWT-based secure login
- Role-based access control

2. CRM Module

- Add/Update/Delete leads
- Customer communication history
- Sales stage transitions

3. Project Management Module

- **Create projects**
- **Assign tasks to employees**
- **Update task progress**

4. Staff & Payroll Module

- Attendance update
- Salary calculation
- Salary slip generation

5. Billing & Invoicing Module

- Create invoices
- Add tax and discount

6. Inventory Module

- **Add items**
- **Update stock movements**
- **Low stock triggers**

7. Analytics Module

- **Real-time business reports**
- **Graphs for revenue, sales, productivity**
- **Downloadable summaries**

The backend follows MVC (Model-View-Controller) principles for clean code structuring.

4.4 Database Implementation

Based on system needs, either MongoDB (NoSQL) or MySQL (Relational) is used.

4.4.1 Key Collections/Tables

USER Table:

- UserID
- Name
- Email
- Password
- Role
- Status

CUSTOMER Table:

- CustomerID
- Name
- Contact
- Lead Stage
- Notes

PROJECT Table:

- ProjectID
- Project Name
- Description
- Deadline

TASK Table:

- TaskID
- AssignedTo

- Status
- Priority

All data is stored with timestamps for reporting purposes.

4.5 Module Integration

Each module communicates through secured REST APIs. Integration Steps:

1. Frontend sends API request
2. Backend validates and processes data
3. Database performs operations
4. Backend returns response to UI
5. UI updates the screen dynamically

This ensures real-time updates for CRM, Billing, Projects, Staff, and Inventory.

4.6 User Interface Screens (Sample Descriptions)

4.6.1 Login Page

- Email/Password input
- Login button
- Validation messages

4.6.2 Admin Dashboard

- Total Users
- Total Invoices
- Projects Summary

4.6.3 CRM Interface

- Customer list with status
- Add Customer button
- Follow-up reminders

4.6.4 Project & Task Management Screen

- Gantt-style view for project timeline
- Task progress percentage
- Team assignment options

4.6.5 Billing Screen

- Invoice form
- Client dropdown
- Auto-generated invoice preview

4.6.6 Inventory Screen

- Item list
- Stock count
- Alerts for low stock

These screens are designed for simplicity and functional clarity.

4.7 Implementation Challenges

Some key challenges encountered during development include:

- Managing multiple modules in a single platform
- Ensuring real-time syncing between CRM, Billing, and Inventory
- Designing a consistent UI for different user roles
- Handling security for sensitive data (payments, employee info)

However, these challenges were resolved through modularization, optimized data flow, and proper backend validation.

Chapter 5 Software Testing

Software testing is a crucial phase in the development of the Trade Track – Intelligent Marketplace Management System, as it ensures that all modules work correctly, meet user requirements, and maintain system reliability. This chapter describes the testing strategies, testing methods, test cases, results, and error resolution followed during development. The goal of testing is to verify that every module—CRM, Billing, Projects, Staff Management, Inventory, and Analytics—operates smoothly and delivers accurate and expected outputs.

5.1 Introduction to Software Testing

Software testing helps identify defects, verify functionality, and ensure that the system performs efficiently under different conditions.

The main objectives of testing Trade Track include:

- Ensuring each feature behaves as expected
- Validating correct data flow between modules
- Confirming UI responsiveness and user experience
- Ensuring security of user and business data
- Verifying system performance and reliability
- Detecting bugs early to reduce maintenance efforts

Trade Track undergoes multiple levels of testing including unit testing, integration testing, system testing, and user acceptance testing.

5.2 Types of Testing Performed

5.2.1 Unit Testing

Unit testing checks individual components like:

- Login module
- CRM lead creation
- Invoice generation
- Task assignment

Each function is tested with valid and invalid data.

5.2.2 Integration Testing

This verifies how well modules interact with each other.

Examples:

- CRM lead converts to customer → Billing module fetches customer
- New project added → Task module receives project ID
- Invoice updated → Reflects in analytics dashboard
- Stock updated → Alerts generated in Inventory module

This ensures consistency in cross-module operations.

5.2.3 System Testing

System testing validates the entire Trade Track system functioning as a whole.

Key aspects tested:

- Complete user workflows
- Dashboard data accuracy
- Multi-user login
- API responses
- System performance and error handling

5.2.4 Functional Testing

Ensures each feature works according to the functional requirements.

Examples:

- Lead should move through CRM stages
- Invoice should auto-calculate totals
- Tasks should change status when updated
- Inventory should reduce when sale happens

5.2.5 Non-Functional Testing

Includes:

- Performance testing – Loading time, API response
- Usability testing – User-friendly UI
- Security testing – Password encryption, protected routes
- Compatibility testing – Works on all browsers/devices

5.2.6 User Acceptance Testing (UAT)

Final testing performed from the user's perspective.

Business owners and employees test:

- CRM workflows
- Billing and staff management
- Inventory checks
- Reports

Feedback is collected and applied for refinements.

5.3 Test Case Design

The following tables describe test cases executed on major modules:

Test Case ID	Description	Input	Expected Output	Status
CRM-01	Add Lead	Customer details	Lead saved successfully	Pass
CRM-02	Update Lead Stage	Move from Lead → Prospect	Stage updated	Pass
CRM-03	Invalid Phone Number	Invalid contact	Error message	Pass
CRM-04	Delete Customer	Customer ID	Customer removed	Pass

5.3.2 CRM Module Test Cases

Test Case ID	Description	Input	Expected Output	Status
CRM-01	Add Lead	Customer details	Lead saved successfully	Pass
CRM-02	Update Lead Stage	Move from Lead → Prospect	Stage updated	Pass
CRM-03	Invalid Phone Number	Invalid contact	Error message	Pass
CRM-04	Delete Customer	Customer ID	Customer removed	Pass

5.3.3 Project & Task Management Test Cases

Test Case ID	Input	Expected Output	Status
PM-01	Create project	Project added	Pass
PM-02	Assign task	Employee receives task	Pass
PM-03	Update status	Task updated to “Completed”	Pass
PM-04	Wrong ID	Invalid task ID	Error displayed

5.3.4 Billing & Invoice Test Cases

Test Case ID	Description	Input	Expected Output	Status
BL-01	Create Invoice	Valid client & amount	Invoice generated	Pass
BL-02	Auto Total	Quantity & rate	Auto-calculation correct	Pass
	Invalid			
BL-03	Amount	Negative values	Error message	Pass
BL-04	Update Invoice	Payment marked as paid	Status updated	Pass

5.3.5 Inventory Module Test Cases

Test Case ID	Description	Input	Expected Output	Status
INV-01	Add Product	Product data	Product saved	Pass
INV-02	Stock Update	Deduct stock	Updated quantity shown	Pass
INV-03	Low Stock	Stock < 10	Low stock alert	Pass
INV-04	Invalid Product	Wrong ID	Error message	Pass

5.4 Bug Reporting & Resolution

Some issues found during testing included:

- Incorrect dashboard count values
- Slow API response during invoice generation
- Missing validation on CRM lead form
- Stock not updating after invoice creation
- Task status not refreshing immediately

Fixes applied:

- **Optimized database queries**
- **Added frontend validations**
- **Improved API pagination**
- **Fixed React state update issues**
- **Added background services for auto-sync**

5.5 Test Results

- **After completing all testing steps:**
- **All major modules are functioning correctly**
- **No critical or high-level defects remain**
- **System is stable, responsive, and secure**
- **User feedback confirms ease of use and clarity**
- Analytics and reporting generate accurate data

The platform meets the functional and non-functional requirements defined earlier.

Chapter 6

Conclusion

The development of Trade Track – An Intelligent Marketplace Management System successfully demonstrates how an integrated, cloudbased platform can simplify and centralize business operations for startups and small-to-medium enterprises (SMEs). The system brings together multiple essential functions—such as Marketing & CRM, Project and Task Management, Staff & Payroll, Billing & Invoicing, Inventory Management, and Analytics—into one unified ecosystem.

Throughout the project, a strong focus was placed on understanding real-world problems faced by SMEs, such as data fragmentation, high subscription costs for multiple tools, inefficient workflows, and lack of real-time insights. By addressing these issues, Trade Track provides a practical and intelligent solution that enhances productivity, reduces manual work, and promotes data-driven decision making.

The iterative design and implementation approach helped ensure modularity, scalability, and maintainability. Comprehensive testing proved that the system operates reliably across all modules and delivers consistent performance. The final system is user-friendly, secure, and accessible across devices through cloud deployment.

In conclusion, Trade Track achieves its goal of offering an affordable, scalable, and efficient business management platform. It demonstrates the effectiveness of combining modern web technologies with practical business workflows, making it a valuable solution for organizations seeking digital transformation and streamlined operations.

Chapter 7

Summary

7.1 Summary

This project presented the design and development of Trade Track – An Intelligent Marketplace Management System, created to centralize business operations into a single platform. The system integrates CRM for customer management, modules for project coordination, automated billing and invoicing, staff and payroll management, and inventory monitoring. Together, these modules help eliminate the need for multiple disconnected tools. The project followed a structured approach beginning with requirement

analysis, system design, implementation, and testing. Key highlights include:

- A unified dashboard enabling real-time visibility across all business functions
- Secure login with role-based access control
- Automated workflows for improved efficiency
- Scalable cloud architecture suitable for growing businesses
- Detailed testing ensuring system reliability and performance

Overall, Trade Track proves to be a robust and user-friendly platform capable of supporting SME operations with ease and intelligence.

7.2 Future Scope

Although the current version of Trade Track provides a wide range of features, several enhancements can further improve performance, scalability, and usability. Future improvements include:

1. Mobile Application (Android & iOS)

Developing a mobile app for quick access to CRM, billing, inventory updates, and task management.

2. Advanced AI-Based Analytics

- Integrating machine learning for:
- Predictive sales forecasting
- Customer churn prediction

- Automated business recommendations

3. Automated Inventory Forecasting

AI models can predict stock requirements and reorder levels based on past usage patterns.

4. Multi-Business Support

Allowing one user to manage multiple businesses/accounts within a single platform.

5. Third-Party Integrations

Supporting integrations with:

- WhatsApp Business API
- Email Marketing Tools
- Payment Gateways
- Accounting software (e.g., Tally, QuickBooks)

6. Voice and Chatbot Assistance

Implementing AI chatbots for task reminders, invoice creation, and customer queries.

7. Multi-Language Support

To make the platform usable by diverse business owners across India and globally.

8. GST Automation

Auto-calculation and filing of GST returns directly through the platform.

9. Role-Based Custom Dashboards

Customization of dashboards based on business requirements and industry types.

7.3 Conclusion of the Report

The project lays a strong foundation for a commercial-grade business management platform. With continuous enhancements and advanced automation, Trade Track has the potential to become a complete enterprise solution capable of supporting large-scale business operations.

References

Key References used:

1) Node.js Documentation - Used to understand server-side JavaScript execution, asynchronous programming, and backend environment setup. <https://nodejs.org/en/docs/>

2) Express.js Documentation - Referenced for creating REST APIs, routing, middleware, request handling, and server configuration.
<https://expressjs.com/>

3) MongoDB Documentation - Consulted for understanding NoSQL database structure, CRUD operations, indexing, and general data storage concepts.
<https://www.mongodb.com/docs/>

4) Mongoose Documentation - Used for designing schemas, connecting Node.js with MongoDB, and performing structured database operations.
<https://mongoosejs.com/docs/>

5) Vite.js Documentation - Referenced for frontend setup, development server configuration, asset bundling, and hot module replacement.
<https://vitejs.dev/guide/>

6) JavaScript (MDN Web Docs) - Used for DOM manipulation, event handling, frontend logic, and browser-based scripting.
<https://developer.mozilla.org/en-US/docs/Web/JavaScript>

7) Git Documentation - Used for version control, repository management, commits, and collaboration.
<https://git-scm.com/doc>
